

Econ490/690: Machine Learning for Finance

Department of Economics, Duke University | Fall 2026

Instructor Information

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Office Hours: See Canvas or by appointment.

Class Schedule and Delivery

Synchronous, In-Person || Social Science 105

24Aug2026–4Dec2026 || Tuesdays and Thursdays || 8:30 AM–9:45 AM

Pre-Requisites

ECON204D - Econometrics and Data Science

Statistics - Working knowledge of statistics and econometrics.

Computing - Some familiarity with a statistical computing language is required. We'll be using R in the course, but familiarity with Python, Julia, Matlab, or a similar language should provide a sufficient foundation to transition to R.

Finance - Some familiarity with investing concepts such as “long,” “short,” “leverage,” and “volatility” is highly recommended.

Course Description

Modern finance analysts must do more than understand statistics, econometrics, or computer science in isolation—they must synthesize these disciplines in applied, data-rich environments. Machine learning sits at this intersection, offering powerful tools to refine traditional financial analysis and unlock new insights from complex, high-dimensional data. This course is designed to help students bridge theory and practice, equipping them with the skills to implement machine learning methods in real-world financial contexts where domain knowledge, statistical rigor, and computational fluency must work together.

This course is structured in four units:

1. Systematic Investment Process
2. Unsupervised Learning
3. Supervised Learning
4. Time Series Analysis

Each unit is motivated by one or more financial economic applications, such as clustering for portfolio construction. For each application, we introduce the statistical and econometric foundations underlying the relevant machine learning method. While we do not linger on formal derivations or proofs, we develop enough mathematical structure to understand the method's intuition, strengths, and limitations. We will then illustrate how to implement these methods using our computational tools. Finally, we will apply them to real data to address the motivating application.

Class sessions are a mixture of lecture and laboratory work. Generally speaking, the first half of each class session develops the relevant concepts and theory in a group setting using lecture slides. The second half of

each class session is spent coding, wherein I will demonstrate key functionality, and you will work to follow along and complete code chunks with classmates.

Communication

Assignments, announcements, grades, readings, and other information will be posted on Canvas. The Canvas site will be the primary method of communication for this course, so please check it frequently.

Course Materials

There is no required textbook for this course. Optional readings will be made available throughout the term. Some initial suggestions include:

- *Hands-On Machine Learning with R*, by Boehmke and Greenwell
- *Econometric Analysis*, by Greene
- *The Econometrics of Financial Markets*, by Campbell, Lo, and MacKinlay
- *Machine Learning in Finance*, by Dixon, Halperin, and Bilokon

Technology

- Cell phones and similar devices: Please do not use cell phones or similar devices during class unless otherwise instructed.
- Laptops and tablets: Please bring your laptop to every class. You are welcome to take notes on a tablet. However, the work we will be doing together is most conducive to a laptop. Specifically, we will be using R regularly. Be sure to have the required R computing environment available on your laptop. Class demonstrations and solutions will generally be completed on Windows, although equivalent workflows are available on macOS.
- Canvas: Course materials such as data, slides, and assignments will be distributed through the Canvas course site.
- 24/7 support is available for Canvas if you encounter technical difficulties. For immediate assistance with other learning technologies, system errors/outages, or your NetID account and password, contact the OIT Service Desk.

Grading

This course centers around a semester-long experiential activity wherein you and a team of classmates run your own systematic, quantitatively focused hedge fund.

Each student's grade is built from a combination of three categories: class participation and two team-based deliverables (Roadshows and Due Diligence Materials). These team-based deliverables account for the vast majority of your course grade.

Each team will receive a score for each team-based deliverable. A student's individual score for that deliverable may, and often will, differ from the team's score depending upon the quantity and quality of the student's individual contributions. These individual contributions will be informed by peer reviews. Moreover, I will make my own assessment of individual contributions based upon attribution within the team deliverables and interactions with students throughout the semester.

I will use the categories outlined below to develop a numeric score. Numeric scores will be translated into letter grades using final course-grade thresholds determined by the instructor, taking into account the overall difficulty and performance of the class.

I will consider regrade requests. Note that during such regrades i) I reevaluate all of the student's submitted materials, not just the assignment in question, and ii) the resulting grade may be higher or lower than the grade assigned prior to the regrade request.

Roadshows (60%): Each team will participate in roughly 5 Roadshows. In the context of our experiential activity, Roadshows are opportunities for a hedge fund to present an overview of its portfolio and investment process for the purpose of raising capital from investors. You have a target cumulative capital raise of \$100Mln. The first four Roadshows focus on "friends and family" money, with a total cap of \$5Mln per Roadshow. The final Roadshow is aimed at institutional investors, with a cap of \$80Mln.

These Roadshows will take place in class. Depending upon class size and scheduling, we might invite outside professionals to serve in the role of investors.

Due Diligence Materials (30%): Within the context of our experiential activity, your fund is invited to provide materials for an institutional investor to conduct a comprehensive due diligence. These materials will provide a comprehensive, professionally presented account of the fund's performance, research process, investment methodology, and evolution over the semester, accompanied by the data and code necessary to reproduce the material analysis.

Class Participation (10%): Participation is judged in several ways.

- **Attendance**: Active participation is essential in this course. If you miss class, you are responsible for staying up to date and reviewing notes from classmates. Absences due to illness, emergency, or university-related obligations are excused when you submit an Incapacitation Form or Dean's Excuse. Excessive absences, excused or unexcused, may impact your participation grade.
- **In-Class Exercises**: Most class sessions will culminate in a hands-on exercise that applies the skills we explore that day. I will guide the class through portions of the exercise and, time permitting, you'll have the opportunity to work in small groups. These submissions are graded for effort and engagement, not for correctness, and you are welcome to submit partial code as long as it reflects a genuine attempt to work through the exercise.

- Engagement: Being engaged with the course material is vital for success. A few examples include: coming to class prepared by completing any necessary readings and pre-work; asking questions in and out of class after having thought carefully about the issue; answering questions I pose to the class; asking your classmates thoughtful questions during presentations; etc.
- Norms: I expect professional behavior and adherence to the Duke Community Standard at all times. Cell phones and similar devices should not be used during class unless explicitly told to do so. Other examples include, but are not limited to, being on time and being respectful of classmates. Failure to adhere to these expectations may adversely affect the Class Participation grade. Serious violations may also be reported to the Office of Student Conduct and Community Standards.

For Master's students, the grading structure is altered slightly:

- Roadshows (50%) – same as above.
- Due Diligence Materials (25%) – same as above.
- Class Participation (10%) – same as above.
- Graduate Research Extension (15%) – an individual research requirement connected to the student's team fund. The purpose of the Graduate Research Extension is to demonstrate additional analytical depth, methodological judgment, and independence. Detailed requirements, grading criteria, and proposal instructions are provided in the Hedge Fund Exercise materials.

Tentative Class Schedule

A detailed course schedule is available on Canvas.

- Week 1: Course introduction and technology stack.
- Weeks 2–3: Signal generation and strategy construction.
- Weeks 4–7: Unsupervised learning, covering techniques such as clustering and dimension reduction.
- Weeks 8–13: Supervised learning, including regression and classification. Topics may include linear regression, ridge regression, LASSO, logistic regression, decision trees, random forests, boosting, Cat-Boost, and support vector machines. We will also examine appropriate tools for model evaluation, such as confusion matrices and ROC curves.
- Weeks 14–15: Time series analysis, including traditional models such as exponential smoothing and ARIMA, along with selected machine-learning approaches to sequential data. Time permitting, we may also discuss neural networks, LSTM models, and related methods.

Other Disclosures:

Modifications: The instructor reserves the right to alter any of the aforementioned components of this syllabus or other class-related materials in an effort to enhance the pedagogical experience.

Salutation: While we are working together in this official capacity, my preferred salutation is Prof. Aguilar. Once our work together is complete, I welcome less formality.

Academic Integrity: I take academic integrity seriously. All students are expected to adhere to the Duke Community Standard. It is the responsibility of each student to understand and follow Duke policies regarding academic integrity. Examples of academic dishonesty include, but are not limited to, cheating on an exam, paraphrasing or summarizing any source without adequate documentation, handing in someone else's work and claiming it is yours (including services such as Coursehero), collaborating with another student inside or outside this class on graded work that is to be done independently, and recycling your own work or the work of other students from other courses without authorization.

Generative AI may not be used for class activities or assignments unless its use is explicitly permitted by the instructor for that activity or assignment.

If I see evidence of any academic dishonesty, I will confront it according to the procedures described in the Bulletin of Duke University.

The Duke Community Standard says:

Duke University is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, and accountability. Citizens of this community commit to reflect upon these principles in all academic and non-academic endeavors, and to protect and promote a culture of integrity. To uphold the Duke Community Standard, students affirm their commitment to uphold the values of the Duke University community by signing a pledge that states:

- I will not lie, cheat, or steal in my academic endeavors.
- I will conduct myself honorably in all my endeavors.
- I will act if the Standard is compromised.

Academic Accommodations: If you are a student with a disability and need accommodations for this class, it is your responsibility to register with the Student Disability Access Office (SDAO) and provide them with documentation of your disability. The SDAO will work with you to determine what accommodations are appropriate for your situation. Please note that accommodations cannot be provided until the instructor receives a Faculty Accommodation Letter from the SDAO. Contact: sdao@duke.edu

Resources to Support Student Success: If you are having trouble completing assignments or understanding the materials, please consult with me about appropriate course preparation and readiness strategies as needed. Either send me an email or visit office hours to discuss the personal or academic difficulties you are facing. I may also direct you to other resources on campus.

The Academic Resource Center (ARC) offers services to support students academically during their undergraduate careers at Duke. The ARC can provide support with time management, academic skills and strategies, course-specific tutoring, ADHD/LD coaching, and more. ARC services are available free to any Duke undergraduate students studying any discipline. Contact: (919) 684-5917 or theARC@duke.edu

Mental Health and Wellness Resources: Student mental health and wellness are of primary importance at Duke, and the university offers resources to support students in managing daily stress and self-care. Some resources are listed below:

DuWell provides Moments of Mindfulness (stress management and resilience building) and meditation programming (Koru workshop) to assist students in developing a daily emotional well-being practice. All are welcome and no experience is necessary.

If your mental health concerns and/or stressful events negatively affect your daily emotional state, academic performance, or ability to participate in your daily activities, many resources are available to help you through difficult times.

DukeReach provides services for students who are experiencing significant challenges relating to mental health, physical health, social adjustment, and/or a variety of other stressors.

Counseling and Psychological Services (CAPS) services include individual and group counseling services, psychiatric services, and workshops. To initiate services, walk-in/call-in 9:00 AM–4:00 PM (M/W/Th/F) and 9:00 AM–6:00 PM Tuesdays. CAPS also provides referral to off-campus resources for specialized care. Contact: (919) 660-1000.

TimelyCare (formerly known as Blue Devils Care) is an online platform that is a convenient, confidential, and free way for Duke students to receive 24/7 mental health support through TalkNow and scheduled counseling.

Financial Support: If you are having difficulty with textbook, software, or supply costs associated with this course, here are some resources:

- Contact the financial aid office, whether you are on aid or not. They have loans and resources for connecting students with programs on campus that might be able to help alleviate these costs.
- DukeLIFE offers course materials assistance for eligible students. Please note that students who are eligible for DukeLIFE benefits are notified prior to the start of the semester; program resources are limited. Students who have limited access to computers may request loaner laptops through the DukeLIFE Technology Assistance Program. Please note that supplies are limited.
- Duke Libraries offers textbook rentals through the Top Textbooks program, where you can rent a textbook for 3 hours at a time.

Discussion Guidelines: Civility is an essential ingredient for academic discourse. Differences in beliefs, opinions, and approaches are to be expected. Active interaction with peers and your instructor is essential to success in this course, so please adhere to these guidelines:

- Respect that others' opinions and beliefs may differ from your own. If you disagree, you may critique the idea, but not the person.
- Listen carefully, be courteous, and do not interrupt.
- Support your statements with evidence and a rationale.
- Try to moderate how you contribute to the discussion—if you have a lot to say, try to avoid dominating the conversation; if you are reluctant to speak up, try to find an opportunity to share your perspective.

Please bring any communications you believe to be in violation of this class policy to the attention of your instructor.